



Usability Test

of the USDA ACIR Portal

Apex Five

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What is ACIR?

The ACIR portal is a USDA database used by importers, customs brokers, and government officials to look up import requirements for agricultural goods

- Supports over 100,000 country-commodity combinations
- Previous research phases (interviews, survey, comparative evaluation) found issues with search, regulatory language, and information organization

USDA Animal and Plant Health Inspection Services
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Agricultural Commodity Import Requirements

APHIS Import Requirements Search Tool for Plant, and Agriculturally Related Products



Plants for Planting and Propagation

Imported material of plants and vegetative parts that are for or capable of propagation, including buds, corms, cuttings, layers, pollen, scions, seeds, tissue, tubers, and like structures



Plants and Plant Products Not for Propagation

Fruits, vegetables, seeds, cut flowers, or handicrafts not for propagation, including products that are fresh, cooked, dried, cut, frozen, and other similar processed states



Wood Products and Byproducts

Logs, lumber, cross ties, firewood, wood chips, mulch, wood packaging material, cork and other similar articles



Pest Organism Permitting

Fungi, insects (including butterflies, bees, and moths), regulated organisms used in pet trade (or smuggled), arthropods (earthworms), parasitic plants, bacteria, noxious weeds and more. See also nonregulated organisms.



Other Items

Tools, materials, or machinery that can act as carriers for pests, including vehicles, containers, bags, packing materials, and other similar articles



Soil, Soil Constituents and Rocks

Compost, manure, rock, sand, soil, and other similar articles



Treatments

Phytosanitary treatments for quarantine-significant plant pests on imported commodities

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USDA.gov

USDA APHIS

Accessibility Statement

USDA Privacy Policy

Non-Discrimination Statement

Information Quality

Whitehouse.gov

FOIA

Plain Writing

USA.gov

Our Methods

- **Research Focus & Justification:** Evaluated the gap between manual lookups and the CIR identification system to address usability issues consistently identified in prior research phases.
- **Participants & Protocol:** Seven regulatory professionals (APHIS, CBP, and private sector) performed moderated "think-aloud" tests via Zoom, following a pilot study that refined task clarity.
- **Tasks**
 - **Task 1 (Search Workflow):** Determine the import status of fresh *Prunus armeniaca* from Mexico to evaluate the core search-to-result process.
 - **Task 2 (Cross-Referencing):** Locate "Small Lots of Seed" in the Plants for Planting Manual for *Solanum lycopersicum* and link it to the correct CIR document to test cross-referencing gaps.
 - **Task 3 (Taxonomy/Complex):** Verify the accepted scientific name for *Leucanthemum nipponicum* and find specific quarantine requirements for Dominica to test synonym resolution.
- **Evaluation:** Sessions concluded with a five-question debriefing to assess task difficulty, label clarity, and participant confidence in document accuracy.

Key Findings

Task 1: Search Workflow

The core process is intuitive for experienced users, but the absence of explicit "Permitted/Prohibited" status banners slows down speed and reduces user confidence.

Task 2: Cross-Referencing

Navigation is described as "taxing" due to a lack of direct hyperlinks and critical gaps where manual references lead to missing documents like "Designated Pest-Free Areas".

Task 3: Taxonomy & Name Resolution

This was the most difficult task because synonym resolution requires too many steps, and search results are often cluttered by irrelevant family-level data.

Recommendations

- **Strengthen System Connections:** Implement direct, functional hyperlinks between the "Plants for Planting" manuals and specific CIR documents to eliminate current navigational gaps.
- **Improve Search Flexibility:** Develop more forgiving search inputs that allow for broad keywords and common names to reduce the reliance on precise scientific naming.
- **Enhance Visual Indicators:** Add explicit "Permitted," "Prohibited," or "Restricted" status banners to CIR pages to provide immediate, high-confidence feedback on admissibility.
- **Increase Resource Visibility:** Improve the discoverability of the Taxonomy page and secondary resources, such as pest-free area lists, to support users who lack deep domain expertise.
- **Optimize Information Hierarchy:** Refine search results to prioritize species-specific data over broad family-level clutter, reducing the cognitive load required to verify requirements.



Usability testing of the USDA ACIR Portal

Usability testing done on three different tasks, each leading to findings

1. Task #1 (Search Workflow) → Intuitive workflow; lacks clear status banners.
2. Task #2 (Cross-Referencing) → Taxing navigation; missing hyperlinks and resources.
3. Task #3 (Taxonomy & Name Resolution) → High cognitive load; cluttered taxonomy results.

Recommendations:

1. **Direct Hyperlinking:** Connect manuals to CIR documents to bridge navigational gaps.
2. **Flexible Search:** Support common names and broad keywords to reduce scientific naming reliance.
3. **Status Banners:** Add explicit "Permitted/Prohibited" labels for immediate admissibility feedback.
4. **Resource Discovery:** Increase visibility of the Taxonomy page and secondary reference lists.
5. **Hierarchy Optimization:** Prioritize species-level results over family-level data to reduce clutter.

THANK YOU

Introduction:

Purpose:

Evaluate how effectively users navigate, interpret, and apply regulatory information in the Agricultural Commodity Import Requirements (ACIR) portal.

Project Focus:

- Assess navigation, search, and filter usability
- Evaluate clarity and organization of help resources
- Examine mobile experience and small-form-factor use cases

Key Research Areas:

- Efficiency of upfront vs. post-search filtering
- User ability to locate commodity-specific requirements
- Effectiveness of current help pathways (labels, videos, guides)
- Opportunities for a centralized “single help hub”



ACIR Search Tools - Locator Table and Frequently Searched

Are you looking for a way to browse topics in ACIR? Would you like a prebuilt search list of Commodity Import Requirements (CIRs) on a specific topic?

There are two Search Tools available in ACIR:

- **Locator Table** - Found on the [ACIR Help Page](#)
 - Allows users to browse various commodities and locate specialty items in ACIR.
 - Topics are listed alphabetically and include Bags/Bagging, Bee Products, Corn, Fresh Fruits and Vegetables, Herbarium Specimens, Lichen, Live Plants, Plant Oils, Seed Kits, Wooden Handicrafts and much more.
- **Frequently Searched** - Found on all search tile pages, above the Search button
 - Allows users to choose prebuilt searches for quick and easy results based on the tile content.
 - [Plants and Plant Products Not for Propagation](#) has the most topics available (e.g., Seeds, Nuts, and Grains - Fresh Fruit and Vegetables from a certain country - Handicrafts - Mushrooms - Tea Products, etc.)
 - Click on the Select button to the right of the topic to view the search results with commodity import requirements related to the topic.

APEX FIVE



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